CRYPTOLOGIC ARCHIVES / COLLECTION 02

International cryptography

Constructions and historical records

Soviet Union and Russia / China / Germany / Japan / Republic of Korea

Research snapshot 2026-10-17

18 profiles; 37 dated milestones; 37 source records in this collection.

Original documents, public specifications, mathematical structure and carefully bounded deductions.

Companion collection: NSA Suite A, preserved separately in the same archive.

Scope and evidence

This collection extends the archive across Soviet/Russian, Chinese, German, Japanese and Korean cryptography. It records public national standards, historical military machines and government systems for which only limited architectural information is public.

Regional labels describe design or historical context. They are not assertions of exclusive use, present-day approval or equivalence to NSA Suite A. Hashes, signatures, key-establishment protocols and encryption ciphers have distinct roles.

Every profile separates source-reported statements, deductions and unresolved details. A patent application, standard issue, implementation date, procurement contract, RFC publication and approval withdrawal are different event types. A public standards document does not identify a country's entire classified cryptographic inventory.

What can be made concrete

Published specifications expose operations such as modular addition, byte substitution, linear diffusion, elliptic-curve point multiplication and rotor permutations. Government product histories sometimes establish only a role, implementation boundary or migration. Each profile states what its cited records actually support.

Source formats and translations

The archive preserves PDFs, original RFC text and raw web snapshots. Each acquired file has a SHA-256, timestamp and original/resolved URL in data/sources.json. English translations, edited transcriptions and scans are distinguished.

The seven hosted Chinese standard translations lack official dated Chinese covers. Their filenames do not independently establish an edition or translation authority. SM2 Part 5 illustrates the issue: the filename contains 2016, while RFC 8998 cites a 2017 standard.

The Fialka manual copy explicitly lacks pages 2, 4 and 60. The Tunny/Newmanry PDF preserves only part of the complete wartime report. These limitations remain visible in the catalogue.

Comparison of constructions

Region	Scheme	Role / construction	Parameters
Soviet Union / Russia	GOST 28147-89	Symmetric encryption and MAC Feistel block cipher	64-bit block; 256-bit key; 32 encryption rounds
Russia	Magma	Symmetric encryption Feistel block cipher	64-bit block; 256-bit key; 32 round keys
Russia	Kuznyechik	Symmetric encryption Substitution and linear-diffusion network	128-bit block; 256-bit key; ten round keys
Russia	Streebog / GOST R 34.11-2012	Cryptographic hashing Iterated compression with substitution, permutation and linear transforms	256- or 512-bit digest; 512-bit chaining state
Russia	GOST R 34.10-2012	Digital signatures Elliptic-curve discrete-logarithm scheme	Prime-field elliptic curves; 512- or 1024-bit encoded signatures
Soviet Union	FIALKA / M-125 family	Historical machine encryption Electromechanical rotor substitution	Ten rotors; 30 contacts per rotor on examined M-125 specimens
China	SM4 / SMS4	Symmetric encryption Unbalanced Feistel-style word transformation	128-bit block; 128-bit key; 32 rounds
China	SM3	Cryptographic hashing Iterated word-oriented compression	256-bit digest; 512-bit message blocks; 64 compression steps
China	SM2 family	Signatures, key exchange and public-key encryption Elliptic-curve cryptography	TLS profile uses a 256-bit prime-field curve
China / international mobile standards	ZUC / Zu Chongzhi	Keystream generation LFSR plus nonlinear state machine	128-bit key; 128-bit IV; 32-bit output words in version 1.6
Germany	Enigma	Historical machine encryption Stepped rotor permutations with a reflector	26-symbol service-machine description; model-dependent rotor/plugboard settings
Germany	Lorenz SZ / Tunny	Historical teleprinter encryption Additive keystream cipher with wheel-controlled state	Five teleprinter bits; five chi wheels, five psi wheels and motor control
Germany	CHIASMUS	Government file encryption and historical SINA use Proprietary symmetric-encryption system; internal construction unresolved here	Key material and format are product-specific in the cited agency account
Germany	LIBELLE in SINA	Government network protection Algorithm implemented inside a hardware security module	Internal primitive parameters not established by the cited BSI records
Japan	Camellia	Symmetric encryption Feistel network with FL/FLINV layers	128-bit block; 128/192/256-bit keys; 18 or 24 rounds
Japan	CLEFIA	Symmetric encryption Four-branch generalized Feistel network	128-bit block; 128/192/256-bit keys; 18/22/26 rounds
Republic of Korea	SEED	Symmetric encryption 16-round Feistel network	128-bit block; 128-bit key
Republic of Korea	ARIA	Symmetric encryption Substitution and involutive linear-diffusion layers	128-bit block; 128/192/256-bit keys; 12/14/16 rounds

Historical milestones

These dates distinguish applications, publications, standards, system development and approval withdrawal. They are not a uniform list of invention or first-approval dates.

Date / subject	Kind / evidence	Interpretation
1918-02-23 Enigma precursor	patent application S054 · Front page	Application/priority milestone for Scherbius's cipher apparatus; not military approval.
1925-07-08 DE416219	patent publication S054 · Front page	Public patent document; distinguish from the 1918 filing.
1940 Enigma	operating manual S038 · Title page	Naval operational instructions in a contemporary original manual.
circa-1940 Enigma	cryptanalytic treatise S039 · Chapter 1 and editorial note	Turing's mathematical and mechanical account; preserved as an edited transcription.
1942-12 Tunny	cryptanalytic program S040 · Section 31A, original p.276	Newman given the task of developing machine-setting methods.
1943-04 Tunny	cryptanalytic equipment S040 · Section 31A, original p.276	Pilot Robinson and Tunny machines arrive at the Newmanry.
1944 Tunny	wartime reference S055 · TUNNY, CHI, PSI and MOTOR-WHEELS entries	Wartime dictionary records wheel terminology and construction.
1945 Tunny	retrospective report S040 · Introduction	Wartime participants record the work; the archived PDF is a 2001 transcription of one portion.
1989 GOST 28147-89	standard edition S020 · Normative reference GOST28147-89	Date of the cited Soviet standard; English RFC follows in 2010.
1998 SEED	reported development S045 · Section 1.2	RFC states development by KISA since 1998.
1999 SINA	system development contract S041 · Development paragraph	BSI commissions secunet; this does not date LIBELLE's first approval.
2000-07-12 Camellia	designer submission S052 · 2000 submission	Early CRYPTREC algorithm specification.
2001 Chiasmus software	reported development S042 · Section 1, p.4	BSI dates Windows program development to 2001.
2001-09-26 Camellia	specification revision S047 · Title and revision record	Designer specification version 2.0.
2003 ARIA	reported development S046 · Section 1	RFC dates development by Korean cryptographers.
2004 ARIA	national standard S046 · Section 1	Korean national-standard milestone reported by the RFC.
2004-04 Camellia	english rfc S043 · Header	RFC 3713 publication.
2005-03-24 Fialka	digitizer introduction S025 · PDF p.2	Dated introduction to the preserved manual copy; original manual date remains unknown.
2005-12 SEED	english rfc S045 · Header	RFC 4269 publication, replacing RFC 4009.
2006-01 SMS4	reported standard issue S053 · Introduction	The 2008 Diffie/Ledin translation reports issue in January 2006.
2007-06-01 CLEFIA	designer specification S049 · Cover	Sony specification version 1.0.
2008-05-15 SMS4	english translation S053 · Cover	Diffie/Ledin translation version 1.03.
2010-03 GOST 28147-89 / ARIA	english rfc S020 · Header; companion RFC S046	RFCs 5830 and 5794; later publications than the original designs.

Date / subject	Kind / evidence	Interpretation
2011-03 CLEFIA	english rfc S044 · Header	RFC 6114 publication.
2011-06-28 ZUC	specification revision S037 · Cover	Version 1.6 of the public ZUC core specification.
2012-03-21 ZUC / SM4 / SM2 / SM3	industry standard release S034 · Announcement	OSCCA releases GM/T 0001-0004, effective on announcement.
2012 GOST hash and signature schemes	standard edition S023 · Introduction; companion signature RFC S024	GOST R 34.11-2012 and GOST R 34.10-2012 editions.
2013-08 GOST hash function	english rfc S023 · Header	RFC 6986 publication.
2013-12 GOST signature algorithm	english rfc S024 · Header	RFC 7091 publication.
2015 Magma / Kuznyechik	standard edition S021 · Introduction	GOST R 34.12-2015 includes both ciphers.
2016-01 SINA	system brochure S048 · Imprint	Date of the preserved BSI system-description edition.
2016-03 Kuznyechik	english rfc S022 · Header	RFC 7801 publication.
2016-08-29 SM4 / SM2 Part 1	national standard issue S035 · Metadata; companion SM2 record S036	Official GB/T issue milestone, separate from implementation date.
2017-03-01 SM4 / SM2 Part 1	national standard implementation S035 · Metadata; companion SM2 record S036	Official implementation date for the cited GB/T records.
2020-09 Magma	english rfc S021 · Header	RFC 8891 publication.
2021-03 SM algorithms with TLS 1.3	protocol profile S026 · Header and Status of This Memo	RFC 8998 describes the combination; it is an Informational Independent Submission.
2022-06-30 Chiasmus	approval withdrawal S042 · Section 1, p.4	BSI withdraws the software's VS-NfD approval; not an algorithm invention date.

GOST 28147-89

Soviet Union / Russia | Symmetric encryption and MAC

A Soviet standard whose public description exposes the round function, key schedule and modes.

Feistel block cipher

64-bit block; 256-bit key; 32 encryption rounds

Documented statements

The encryption core processes 64-bit blocks using a 256-bit key and 32 rounds. Its round transformation combines addition modulo 2^32, eight 4-bit substitutions, an 11-bit cyclic rotation and XOR.

INTL-GOST-28147-F1 | S020: Sections 4 and 5.1

The 1989 standard leaves substitution tables to a parameter set; RFC 4357 supplies named parameter sets and identifiers.

INTL-GOST-28147-F2 | S020: Section 4, S-box remark and long-term parameter discussion

RFC 4357 gives concrete encryption parameter-set encodings and S-box tables.

INTL-GOST-28147-F3 | S050: Sections 8.1, 11.1 and 11.2

The reference standard is dated 1989; the preserved English RFC was published in March 2010.

INTL-GOST-28147-F4 | S020: Header and Section 10, GOST28147-89 reference

Technical inferences

Strong — The label GOST 28147-89 is insufficient for interoperable encryption: the S-box parameter set and mode must also be specified.

INTL-GOST-28147-I1 | Basis: INTL-GOST-28147-F1, INTL-GOST-28147-F2, INTL-GOST-28147-F3

Unresolved details

The 1989 standard date does not establish the original internal design date or approval for every later product.

Magma

Russia | Symmetric encryption

The updated 64-bit cipher in GOST R 34.12-2015, with important compatibility details.

Feistel block cipher

64-bit block; 256-bit key; 32 round keys

Documented statements

Magma uses a 64-bit block and 256-bit key; its specified encryption composition uses 32 round keys.

INTL-MAGMA-F1 | S021: Abstract; Sections 4.2, 4.3 and 5.1

Relative to GOST 28147-89, the RFC fixes the S-box set and changes key/block parsing conventions, including big-endian key parsing.

INTL-MAGMA-F2 | S021: Appendix B, Background

Its national-standard edition is 2015; RFC 8891 was published in September 2020.

INTL-MAGMA-F3 | S021: Header, Introduction and normative reference GOSTR3412-2015

Technical inferences

Strong — Replacing a legacy GOST implementation by changing its displayed algorithm name is insufficient: representation and parameter choices can change ciphertext.

INTL-MAGMA-I1 | Basis: INTL-MAGMA-F1, INTL-MAGMA-F2

Unresolved details

The 2020 RFC date is an English-publication milestone, not the cipher invention date.

Kuznyechik

Russia | Symmetric encryption

The 128-bit cipher standardized alongside Magma in GOST R 34.12-2015.

Substitution and linear-diffusion network

128-bit block; 256-bit key; ten round keys

Documented statements

The standard specifies a 128-bit block and 256-bit key. Ten round keys are derived, and encryption applies nine keyed substitution/linear-transform layers followed by a final key XOR.

INTL-KUZNYECHIK-F1 | S022: Abstract; Sections 4.4 and 4.5.1

Its nonlinear byte substitution, linear transformation and key schedule are explicitly specified with worked examples.

INTL-KUZNYECHIK-F2 | S022: Sections 4.1-4.4 and 5

The Russian standard is the 2015 edition; the English RFC dates to March 2016.

INTL-KUZNYECHIK-F3 | S022: Header and normative reference GOSTR3412-2015

Technical inferences

Strong — The published transformations and vectors permit independent implementations; implementation agreement can be checked without knowing unpublished round details.

INTL-KUZNYECHIK-I1 | Basis: INTL-KUZNYECHIK-F1, INTL-KUZNYECHIK-F2

Unresolved details

Sharing a standard with Magma does not make the two ciphers mathematically interchangeable.

Streebog / GOST R 34.11-2012

Russia | Cryptographic hashing

A hash-function standard, with a fully specified compression construction.

Iterated compression with substitution, permutation and linear transforms

256- or 512-bit digest; 512-bit chaining state

Documented statements

GOST R 34.11-2012 defines 256-bit and 512-bit digest variants. Both use a 512-bit chaining state and specified S, P and L transformations.

INTL-STREEBOG-F1 | S023: Sections 5-9

The compression function combines a keyed internal transformation with the previous chaining value and message block by XOR; finalization also processes message length and a block sum.

INTL-STREEBOG-F2 | S023: Sections 8 and 9

The standard edition is 2012; the preserved English publication is RFC 6986 of August 2013.

INTL-STREEBOG-F3 | S023: Header and normative reference GOST3411-2012

Technical inferences

Strong — A 256-bit digest is an output length. It is not a 256-bit encryption key, and a hash interface does not provide a decryption operation.

INTL-STREEBOG-I1 | Basis: INTL-STREEBOG-F1, INTL-STREEBOG-F2

Unresolved details

This profile records the specified construction, not a complete survey of subsequent cryptanalysis.

GOST R 34.10-2012

Russia | Digital signatures

The GOST signature standard is a separate mathematical object from GOST block ciphers.

Elliptic-curve discrete-logarithm scheme

Prime-field elliptic curves; 512- or 1024-bit encoded signatures

Documented statements

The standard defines signing and verification using an elliptic-curve point group over a prime finite field. It specifies signature encodings of 512 or 1024 bits.

INTL-GOST-SIGNATURES-F1 | S024: Sections 1.2, 4 and 5

The signing rule uses an ephemeral scalar k, point C=kP, r=x_C mod q, and s=(r*d+k*e) mod q, with the specified hash-to-integer and retry rules.

INTL-GOST-SIGNATURES-F2 | S024: Section 6.1

The standard edition is 2012 and the English RFC is dated December 2013.

INTL-GOST-SIGNATURES-F3 | S024: Header and Section 1.1

Technical inferences

Strong — GOST can name an ecosystem containing signatures, hashes and ciphers. A signature length alone cannot identify the block cipher used by a product.

INTL-GOST-SIGNATURES-I1 | Basis: INTL-GOST-SIGNATURES-F1, INTL-GOST-SIGNATURES-F2

Unresolved details

A product profile must additionally identify curve parameters, hash function and encoding conventions.

FIALKA / M-125 family

Soviet Union | Historical machine encryption

A Cold War cipher-machine family documented through surviving artifacts and a German-language manual.

Electromechanical rotor substitution

Ten rotors; 30 contacts per rotor on examined M-125 specimens

Documented statements

The examined M-125 machines have ten 30-contact rotors; adjacent rotors move in opposite directions. A punched card sets internal connections, and the machine prints and punches paper tape.

INTL-FIALKA-F1 | S051: General Description; Input and Output; Card Reader; Direction of Rotor Rotation

The preserved manual covers M-125-MN and M-125-3MN. Its digitizer introduction is dated 24 March 2005 and explicitly notes missing original pages 2, 4 and 60.

INTL-FIALKA-F2 | S025: PDF pp.1-2, digitizer cover and introduction

Technical inferences

Strong — These are changing symbol permutations implemented with physical contacts and stepping mechanics. A modern fixed-width block-cipher API would omit important machine-state and operating-procedure details.

INTL-FIALKA-I1 | Basis: INTL-FIALKA-F1

Unresolved details

The original manual date and first service-approval date are not verified here.

Variant-specific reflector connections and rotor sets require examination of the particular model.

SM4 / SMS4

China | Symmetric encryption

A public block cipher whose earlier name SMS4 and later standard numbers describe different historical milestones.

Unbalanced Feistel-style word transformation

128-bit block; 128-bit key; 32 rounds

Documented statements

The published transformation uses four 32-bit state words, 32 rounds, byte S-box substitution and a linear layer built from XOR and rotations. Both block and key are 128 bits.

INTL-SM4-F1 | S027: Sections 4, 6 and 7

In compact notation, the update is X[i+4]=X[i] XOR T(X[i+1] XOR X[i+2] XOR X[i+3] XOR rk[i]); output reverses the final four words.

INTL-SM4-F2 | S027: Sections 6.1 and 7.1

Diffie and Ledin's dated May 2008 translation reports that SMS4 was issued in January 2006. OSCCA explicitly identified SM4 as formerly SMS4 in its March 2012 announcement.

INTL-SM4-F3 | S053: Cover and introductory paragraph; S034: Announcement list, GM/T0002-2012

The official national record gives GB/T 32907-2016 an issue date of 29 August 2016 and an implementation date of 1 March 2017.

INTL-SM4-F4 | S035: Publication and implementation metadata

Technical inferences

Strong — A protocol entry such as SM4-GCM identifies a block cipher plus a mode. Its authentication and nonce rules belong to the full protocol profile, rather than to the SM4 round function alone.

INTL-SM4-I1 | Basis: INTL-SM4-F1, INTL-SM4-F2

Unresolved details

The hosted English translation lacks an official dated Chinese cover; its edition is not established by its filename.

SM3

China | Cryptographic hashing

A public hash specification built from Boolean functions, word additions, rotations and message expansion.

Iterated word-oriented compression

256-bit digest; 512-bit message blocks; 64 compression steps

Documented statements

SM3 pads messages into 512-bit blocks and updates a 256-bit chaining value with a compression function. That function iterates j from 0 to 63.

INTL-SM3-F1 | S028: Sections 5.1-5.3.3

The compression state comprises eight 32-bit words, using specified Boolean functions, rotations, additions and permutations P0/P1.

INTL-SM3-F2 | S028: Sections 3, 4 and 5.3.3

OSCCA announced GM/T 0004-2012 for SM3 on 21 March 2012. RFC 8998 later specifies SM3 in its SM TLS 1.3 profile.

INTL-SM3-F3 | S034: Announcement list, GM/T0004-2012; S026: Sections 1.1 and 3.4

Technical inferences

Strong — The concrete object is a digest computation, with message expansion and compression, rather than a bulk-encryption key schedule.

INTL-SM3-I1 | Basis: INTL-SM3-F1, INTL-SM3-F2

Unresolved details

A 256-bit digest length alone does not establish every security property or interoperability with other 256-bit hashes.

SM2 family

China | Signatures, key exchange and public-key encryption

SM2 names several related algorithms; the operation and protocol must be identified.

Elliptic-curve cryptography

TLS profile uses a 256-bit prime-field curve

Documented statements

The preserved parts separately describe general elliptic-curve operations, signatures, key exchange, public-key encryption and parameter definition.

INTL-SM2-F1 | S029: Title and Scope; companion parts S030-S033; S030: Title and Scope; companion parts S031-S033

The signature algorithm hashes an identity-bound value with the message; its signing equations include r=(e+x1) mod n and s=(1+d)^(-1)*(k-r*d) mod n, with required retry conditions.

INTL-SM2-F2 | S030: Sections 5.5 and 6.1

RFC 8998 specifies a 256-bit prime-field curve and uses SM2 signatures with SM3. Its TLS key exchange is ECDHE over that curve.

INTL-SM2-F3 | S026: Sections 1, 3.2.1 and 3.3

OSCCA announced the GM/T SM2 family in March 2012. The official GB/T Part 1 record was issued in August 2016 and implemented in March 2017.

INTL-SM2-F4 | S036: Publication and implementation metadata; S034: Announcement list, GM/T0003-2012

Technical inferences

Strong — Using the SM2 curve for ECDHE in TLS does not by itself mean that the native SM2 Part 3 key-exchange protocol is being executed.

INTL-SM2-I1 | Basis: INTL-SM2-F1, INTL-SM2-F3

Unresolved details

The English Part 5 filename says 2016, whereas RFC 8998 cites GB/T 32918.5-2017. The filename must not be treated as authoritative edition metadata.

ZUC / Zu Chongzhi

China / international mobile standards | Keystream generation

A public word-oriented stream cipher used as a building block for mobile confidentiality and integrity algorithms.

LFSR plus nonlinear state machine

128-bit key; 128-bit IV; 32-bit output words in version 1.6

Documented statements

Version 1.6 takes a 128-bit key and 128-bit IV and produces 32-bit keystream words. The design combines a 16-cell, 31-bit LFSR, bit reorganization and a nonlinear function.

INTL-ZUC-F1 | S037: Sections 1, 2.1-2.4

The preserved specification is dated 28 June 2011 and identifies ZUC as the core for 128-EEA3 and 128-EIA3.

INTL-ZUC-F2 | S037: Cover and Section 1

OSCCA included the Zu Chongzhi industry standard in its 21 March 2012 announcement.

INTL-ZUC-F3 | S034: Announcement list, GM/T0001-2012

Technical inferences

Strong — The keystream generator, the confidentiality algorithm and the integrity algorithm occupy different interfaces; the name ZUC alone does not specify the full mobile packet-protection procedure.

INTL-ZUC-I1 | Basis: INTL-ZUC-F1, INTL-ZUC-F2

Unresolved details

Parameters in this profile apply to the preserved 128-bit version, not automatically to later ZUC-256 variants.

Enigma

Germany | Historical machine encryption

A machine family whose operating procedures and reflector structure are documented in wartime sources.

Stepped rotor permutations with a reflector

26-symbol service-machine description; model-dependent rotor/plugboard settings

Documented statements

Turing describes 26-contact rotors, paired reflector contacts, turnover mechanics and plugboard changes. At a fixed machine position, encryption is reciprocal and no letter maps to itself.

INTL-ENIGMA-F1 | S039: Chapter 1, printed pp.1-4, especially p.2

The 1940 naval manual records a contemporary operational procedure; it is a separate kind of evidence from a patent or a later cryptanalytic account.

INTL-ENIGMA-F2 | S038: Title page and operational instructions

Scherbius's patent record distinguishes its 23 February 1918 application/priority date from publication on 8 July 1925.

INTL-ENIGMA-F3 | S054: Patent front page; S056: Opening paragraph, Scherbius patent application

Technical inferences

Strong — Writing the fixed-position wiring as E=T^(-1)*R*T shows why E is its own inverse when the reflector R is an involution. A reflector without fixed points also makes E fixed-point-free.

Assumption: T includes all forward wiring and the reflector pairs distinct contacts; composition is evaluated at one fixed stepping state.

INTL-ENIGMA-I1 | Basis: INTL-ENIGMA-F1

Unresolved details

The 1918 patent milestone does not establish the service-approval date of every later Enigma model.

Different Enigma variants and operating procedures require separate profiles.

Lorenz SZ / Tunny

Germany | Historical teleprinter encryption

Tunny is the British cryptanalytic name used in the preserved wartime record.

Additive keystream cipher with wheel-controlled state

Five teleprinter bits; five chi wheels, five psi wheels and motor control

Documented statements

The wartime dictionary describes five regularly moving chi wheels, five psi wheels with controlled movement, and motor-wheel control of the psi movement.

INTL-LORENZ-F1 | S055: Entries CHI, PSI, MOTOR-WHEELS and TUNNY

The Newmanry report describes machine-based setting work beginning in December 1942 and pilot Robinson and Tunny machines arriving in April 1943.

INTL-LORENZ-F2 | S040: Section 31A, original p.276

The report treats five-impulse teleprinter characters, chi/psi components and addition modulo 2 in its account of Tunny and decoding machines.

INTL-LORENZ-F3 | S040: Section 56J, original pp.376-378

Technical inferences

Strong — At the additive interface C=P XOR K, reuse of the same keystream yields C1 XOR C2=P1 XOR P2. This is a property of keystream reuse, independent of the number of wheels.

Assumption: The two messages use the same keystream at aligned bit positions.

INTL-LORENZ-I1 | Basis: INTL-LORENZ-F1, INTL-LORENZ-F3

Unresolved details

The preserved 1942-1945 British milestones do not date the original German invention or every SZ40/SZ42 deployment.

The Newmanry PDF is a transcription of part of the report, not the complete wartime original.

CHIASMUS

Germany | Government file encryption and historical SINA use

A German government example where public lifecycle evidence is clearer than the primitive's internals.

Proprietary symmetric-encryption system; internal construction unresolved here

Key material and format are product-specific in the cited agency account

Documented statements

BSI dates the Windows Chiasmus file-encryption program to 2001, followed by a Linux command-line version. Partners had to exchange secret key material beforehand.

INTL-CHIASMUS-F1 | S042: Section 1, printed p.4

BSI describes non-open algorithms and standards and proprietary software; it says Chiasmus key material is incompatible with replacement products.

INTL-CHIASMUS-F2 | S042: Section 1, printed p.4

BSI states that the VS-NfD approval was withdrawn on 30 June 2022. Its SINA history separately records replacing CHIASMUS by AES at lower classification levels.

INTL-CHIASMUS-F3 | S042: Section 1, printed p.4; S041: Cryptographic migration paragraph

Technical inferences

Strong — A file archive or migration tool needs the actual container, key format and implementation behavior; merely selecting another symmetric cipher cannot make a Chiasmus file readable.

INTL-CHIASMUS-I1 | Basis: INTL-CHIASMUS-F1, INTL-CHIASMUS-F2

Unresolved details

The cited agency records do not expose a complete round function or establish the cipher's original design date.

Independent reverse engineering should be catalogued separately from an official design release.

LIBELLE in SINA

Germany | Government network protection

A named government cipher with public architectural and migration evidence.

Algorithm implemented inside a hardware security module

Internal primitive parameters not established by the cited BSI records

Documented statements

BSI's SINA history says development was commissioned from secunet in 1999. It identifies LIBELLE as an HSM-implemented algorithm used at higher classification levels and describes gradual replacement by a newly approved algorithm.

INTL-LIBELLE-F1 | S041: SINA im Wandel der Zeit, development and cryptographic migration paragraphs

The preserved SINA brochure is a January 2016 system description covering protected network communication, storage and separation of functions.

INTL-LIBELLE-F2 | S048: Sections 1 and 4; imprint, January 2016

Technical inferences

Strong — A stable product-family name such as SINA can span changes in cipher and security hardware. The algorithm, hardware module and product version must be identified separately.

INTL-LIBELLE-I1 | Basis: INTL-LIBELLE-F1, INTL-LIBELLE-F2

Unresolved details

The cited sources do not identify the replacement cipher or provide LIBELLE's round function.

The 1999 SINA contract is a system-development milestone, not a verified first approval of LIBELLE.

Camellia

Japan | Symmetric encryption

A designer-published cipher from NTT and Mitsubishi Electric with early CRYPTREC records.

Feistel network with FL/FLINV layers

128-bit block; 128/192/256-bit keys; 18 or 24 rounds

Documented statements

Camellia was jointly developed by NTT and Mitsubishi Electric. It has a 128-bit block and 128-, 192- and 256-bit key options.

INTL-CAMELLIA-F1 | S043: Abstract and Section 1.1

The data path has 18 rounds for 128-bit keys and 24 rounds for the larger keys, with FL and FLINV functions inserted every six rounds.

INTL-CAMELLIA-F2 | S043: Sections 2.3 and 2.4

The designer archive identifies a 12 July 2000 CRYPTREC specification and a 26 September 2001 version 2.0. The English RFC followed in April 2004.

INTL-CAMELLIA-F3 | S047: Version 2.0 title and revision record; companion 2000 submission S052; S052: 2000 designer submission; S043: Header

Technical inferences

Strong — A 128-bit block interface with several key sizes does not imply AES internals: Camellia's specified data path is a Feistel construction with additional keyed layers.

INTL-CAMELLIA-I1 | Basis: INTL-CAMELLIA-F1, INTL-CAMELLIA-F2

Unresolved details

A submission, an algorithm publication and a national procurement approval are separate historical milestones.

CLEFIA

Japan | Symmetric encryption

Sony's published cipher gives an explicit alternative to a two-branch Feistel network.

Four-branch generalized Feistel network

128-bit block; 128/192/256-bit keys; 18/22/26 rounds

Documented statements

Sony's version 1.0 specification is dated 1 June 2007. It uses four 32-bit data branches and 18, 22 or 26 rounds for the three key sizes.

INTL-CLEFIA-F1 | S049: Cover; Sections 1 and 4.2

The data-processing part uses a four-branch generalized Feistel network; an eight-branch construction also appears in scheduling larger keys.

INTL-CLEFIA-F2 | S044: Section 4

The English RFC specification was published in March 2011 and dates the algorithm's public publication to 2007.

INTL-CLEFIA-F3 | S044: Header and Introduction

Technical inferences

Strong — The four-branch structure is a concrete design distinction, despite sharing a 128-bit block interface and key-size choices with several other ciphers.

INTL-CLEFIA-I1 | Basis: INTL-CLEFIA-F1, INTL-CLEFIA-F2

Unresolved details

The preserved design dates do not establish when every Sony product first deployed CLEFIA.

SEED

Republic of Korea | Symmetric encryption

A Korean national cipher with a public description of its round structure.

16-round Feistel network

128-bit block; 128-bit key

Documented statements

RFC 4269 describes SEED as developed by KISA since 1998 and as a Korean national standard. It specifies a 128-bit block, a 128-bit key and 16 Feistel rounds.

INTL-SEED-F1 | S045: Section 1.2

The public round description includes S-box transformations, permutations, XOR and addition, with an explicit key schedule and test data.

INTL-SEED-F2 | S045: Sections 2 and 3; appendices

The preserved RFC is dated December 2005 and replaces the earlier RFC 4009 description.

INTL-SEED-F3 | S045: Header and Abstract

Technical inferences

Strong — SEED provides an example where both the primitive's structure and development period can be stated from a published specification, while product certification remains a separate question.

INTL-SEED-I1 | Basis: INTL-SEED-F1, INTL-SEED-F2

Unresolved details

This profile concerns the 128-bit SEED specification in RFC 4269, not every later similarly named variant.

ARIA

Republic of Korea | Symmetric encryption

A Korean cipher with public development and standardization milestones.

Substitution and involutive linear-diffusion layers

128-bit block; 128/192/256-bit keys; 12/14/16 rounds

Documented statements

RFC 5794 dates development by Korean cryptographers to 2003 and Korean standardization to 2004. It specifies 128-bit blocks with 12, 14 or 16 rounds depending on key size.

INTL-ARIA-F1 | S046: Section 1

The round transformations alternate substitution layers, use XOR key addition and a diffusion map A satisfying A(A(x))=x.

INTL-ARIA-F2 | S046: Sections 2.4.1-2.4.3

The preserved English RFC was published in March 2010.

INTL-ARIA-F3 | S046: Header

Technical inferences

Strong — The diffusion map's involution is a specific algebraic property. It does not mean encryption is self-inverse without the specified key-schedule and round transformations.

INTL-ARIA-I1 | Basis: INTL-ARIA-F1, INTL-ARIA-F2

Unresolved details

The 2004 national-standard milestone is not proof of universal approval for all Korean government information.

Source catalogue

This catalogue covers the international collection. The repository catalogue additionally includes the earlier Suite A sources. Files retain their original notices.

S020 · RFC 5830: GOST 28147-89: Encryption, Decryption, and MAC Algorithms

RFC Editor; named document authors

Document date: 2010-03 | Language: English | Format: text

Provenance: RFC Editor publication; original national standards and RFC dates are distinguished

Preserved in the original publisher format. RFC publication does not itself establish national approval or IETF endorsement; consult the document status statement.

https://www.rfc-editor.org/rfc/rfc5830.txt

Preserved: sources/originals/S020-rfc5830.txt | 36,234 bytes

Retrieved UTC: 2026-10-17T18:50:42.729077+00:00

SHA-256: 5b85a3587130cd2c43e6b3a4a41f6b3bdaa3b320c84e0298a09c4fe1016b663e

S021 · RFC 8891: GOST R 34.12-2015: Block Cipher Magma

RFC Editor; named document authors

Document date: 2020-09 | Language: English | Format: pdf

Provenance: RFC Editor publication; original national standards and RFC dates are distinguished

Preserved in the original publisher format. RFC publication does not itself establish national approval or IETF endorsement; consult the document status statement.

https://www.rfc-editor.org/rfc/rfc8891.pdf

Preserved: sources/originals/S021-rfc8891.pdf | 154,889 bytes | 13 pages

Retrieved UTC: 2026-10-17T18:50:42.731177+00:00

SHA-256: efd5cdbc8335fa7cb5d214cf49b5064c7600a41e9e0430238b71ba3a87f59797

S022 · RFC 7801: GOST R 34.12-2015: Block Cipher Kuznyechik

RFC Editor; named document authors

Document date: 2016-03 | Language: English | Format: text

Provenance: RFC Editor publication; original national standards and RFC dates are distinguished

Preserved in the original publisher format. RFC publication does not itself establish national approval or IETF endorsement; consult the document status statement.

https://www.rfc-editor.org/rfc/rfc7801.txt

Preserved: sources/originals/S022-rfc7801.txt | 25,039 bytes

Retrieved UTC: 2026-10-17T18:50:42.740936+00:00

SHA-256: 4e78cccae8ba582fa2e316aa8cbbb404ae0355d0a06b76d685482c0a0a842c69

S023 · RFC 6986: GOST R 34.11-2012: Hash Function

RFC Editor; named document authors

Document date: 2013-08 | Language: English | Format: text

Provenance: RFC Editor publication; original national standards and RFC dates are distinguished

Preserved in the original publisher format. RFC publication does not itself establish national approval or IETF endorsement; consult the document status statement.

https://www.rfc-editor.org/rfc/rfc6986.txt

Preserved: sources/originals/S023-rfc6986.txt | 78,975 bytes

Retrieved UTC: 2026-10-17T18:50:42.764623+00:00

SHA-256: fd5ea9e36d74743bbc49df7652e82d00aa97195aa40b6594d39ef4f0028e2226

S024 · RFC 7091: GOST R 34.10-2012: Digital Signature Algorithm

RFC Editor; named document authors

Document date: 2013-12 | Language: English | Format: text

Provenance: RFC Editor publication; original national standards and RFC dates are distinguished

Preserved in the original publisher format. RFC publication does not itself establish national approval or IETF endorsement; consult the document status statement.

https://www.rfc-editor.org/rfc/rfc7091.txt

Preserved: sources/originals/S024-rfc7091.txt | 39,924 bytes

Retrieved UTC: 2026-10-17T18:50:43.120820+00:00

SHA-256: 559d87823305916db8996dad7bd954c6c8db58cd34df11748f4f0947b01c5758

S025 · FIALKA M-125-MN / M-125-3MN German-language service manual

Historical service manual; preserved by the Enigma Museum

Document date: not verified | Language: German | Format: pdf

Provenance: Original manual scan preserved by the Enigma Museum; contributed by collector John Alexander

Digitized in 2005; original manual date not verified. The digitizer explicitly identifies original pages 2, 4 and 60 as missing. This copy is preserved unchanged with that limitation.

https://enigmamuseum.com/mfman.pdf

Preserved: sources/originals/S025-fialka-german-manual.pdf | 3,160,433 bytes | 112 pages

Retrieved UTC: 2026-10-17T18:50:43.308709+00:00

SHA-256: 09a7e4fd862dd08154fae39205887e822435c2970dcda009de274922770c6c64

S026 · RFC 8998: ShangMi (SM) Cipher Suites for TLS 1.3

RFC Editor; named document authors

Document date: 2021-03 | Language: English | Format: pdf

Provenance: RFC Editor publication; original national standards and RFC dates are distinguished

Preserved in the original publisher format. RFC publication does not itself establish national approval or IETF endorsement; consult the document status statement.

https://www.rfc-editor.org/rfc/rfc8998.pdf

Preserved: sources/originals/S026-rfc8998.pdf | 189,766 bytes | 13 pages

Retrieved UTC: 2026-10-17T18:50:43.324478+00:00

SHA-256: 6a8296ee227a22e5fdc8551e99833181bc206c79f949809506df5d6b24fbfecd

S027 · GB/T 32907-2016: SM4 block cipher algorithm, English translation

Chinese national standard; translation preserved by Alipay

Document date: not verified | Language: English | Format: pdf

Provenance: English translation preserved in Alipay's public TLS specification repository

The host filename identifies a 2016 GB/T standard; the English translation itself lacks the dated official Chinese cover. Translation edition and authorization are unverified. Official metadata and the 2012 OSCCA announcement are cited separately for standardization dates.

https://raw.githubusercontent.com/alipay/tls13-sm-spec/master/sm-en-pdfs/sm4/GBT.32907-2016.SM4-en.pdf

Preserved: sources/originals/S027-sm4-2016-english.pdf | 150,643 bytes | 8 pages

Retrieved UTC: 2026-10-17T18:50:43.433810+00:00

SHA-256: c7c2570f1a76ab4d1361bc487fb87ab58de659f3054f3fbb6f9e2ba014f85142

S028 · GB/T 32905-2016: SM3 cryptographic hash algorithm, English translation

Chinese national standard; translation preserved by Alipay

Document date: not verified | Language: English | Format: pdf

Provenance: English translation preserved in Alipay's public TLS specification repository

The host filename identifies a 2016 GB/T standard; the English translation itself lacks the dated official Chinese cover. Translation edition and authorization are unverified. Official metadata and the 2012 OSCCA announcement are cited separately for standardization dates.

https://raw.githubusercontent.com/alipay/tls13-sm-spec/master/sm-en-pdfs/sm3/GBT.32905-2016.SM3-en.pdf

Preserved: sources/originals/S028-sm3-2016-english.pdf | 965,071 bytes | 14 pages

Retrieved UTC: 2026-10-17T18:50:43.577123+00:00

SHA-256: 4619044302bd407d9115091b0aee9afa8647c8e6a805ae65e5d22178c013ac82

S029 · SM2 national standard, Part 1, English translation

Chinese national standard; translation preserved by Alipay

Document date: not verified | Language: English | Format: pdf

Provenance: English translation preserved in Alipay's public TLS specification repository

The host filename contains GB/T and 2016, but the English cover lacks a dated official standard identifier. In particular, do not infer a 2016 issue date for Part 5 from the filename: RFC 8998 cites GB/T 32918.5-2017. Translation edition and authorization are unverified.

https://raw.githubusercontent.com/alipay/tls13-sm-spec/master/sm-en-pdfs/sm2/GBT.32918.1-2016.SM2-en.pdf

Preserved: sources/originals/S029-sm2-part1-english.pdf | 1,252,502 bytes | 60 pages

Retrieved UTC: 2026-10-17T18:50:43.749867+00:00

SHA-256: 8bd9ad8f4ae0190837e6ca1ccb9541a1c14f733010c2bf401f4f44e506149500

S030 · SM2 national standard, Part 2, English translation

Chinese national standard; translation preserved by Alipay

Document date: not verified | Language: English | Format: pdf

Provenance: English translation preserved in Alipay's public TLS specification repository

The host filename contains GB/T and 2016, but the English cover lacks a dated official standard identifier. In particular, do not infer a 2016 issue date for Part 5 from the filename: RFC 8998 cites GB/T 32918.5-2017. Translation edition and authorization are unverified.

https://raw.githubusercontent.com/alipay/tls13-sm-spec/master/sm-en-pdfs/sm2/GBT.32918.2-2016.SM2-en.pdf

Preserved: sources/originals/S030-sm2-part2-english.pdf | 685,148 bytes | 14 pages

Retrieved UTC: 2026-10-17T18:50:43.858393+00:00

SHA-256: d2607609564e2d04a242675769bde2cb394b0492d7c52a9d91b50ba858747fde

S031 · SM2 national standard, Part 3, English translation

Chinese national standard; translation preserved by Alipay

Document date: not verified | Language: English | Format: pdf

Provenance: English translation preserved in Alipay's public TLS specification repository

The host filename contains GB/T and 2016, but the English cover lacks a dated official standard identifier. In particular, do not infer a 2016 issue date for Part 5 from the filename: RFC 8998 cites GB/T 32918.5-2017. Translation edition and authorization are unverified.

https://raw.githubusercontent.com/alipay/tls13-sm-spec/master/sm-en-pdfs/sm2/GBT.32918.3-2016.SM2-en.pdf

Preserved: sources/originals/S031-sm2-part3-english.pdf | 740,552 bytes | 23 pages

Retrieved UTC: 2026-10-17T18:50:44.044137+00:00

SHA-256: ecd0384cc479abc0df9651173bc62866c31929140889536cc6c231ffb9a99f5c

S032 · SM2 national standard, Part 4, English translation

Chinese national standard; translation preserved by Alipay

Document date: not verified | Language: English | Format: pdf

Provenance: English translation preserved in Alipay's public TLS specification repository

The host filename contains GB/T and 2016, but the English cover lacks a dated official standard identifier. In particular, do not infer a 2016 issue date for Part 5 from the filename: RFC 8998 cites GB/T 32918.5-2017. Translation edition and authorization are unverified.

https://raw.githubusercontent.com/alipay/tls13-sm-spec/master/sm-en-pdfs/sm2/GBT.32918.4-2016.SM2-en.pdf

Preserved: sources/originals/S032-sm2-part4-english.pdf | 621,510 bytes | 17 pages

Retrieved UTC: 2026-10-17T18:50:44.175444+00:00

SHA-256: ab1f8e93df78bad3fb4db8456feb655e3a660fb12c352638bba3412553131043

S033 · SM2 national standard, Part 5, English translation

Chinese national standard; translation preserved by Alipay

Document date: not verified | Language: English | Format: pdf

Provenance: English translation preserved in Alipay's public TLS specification repository

The host filename contains GB/T and 2016, but the English cover lacks a dated official standard identifier. In particular, do not infer a 2016 issue date for Part 5 from the filename: RFC 8998 cites GB/T 32918.5-2017. Translation edition and authorization are unverified.

https://raw.githubusercontent.com/alipay/tls13-sm-spec/master/sm-en-pdfs/sm2/GBT.32918.5-2016.SM2-en.pdf

Preserved: sources/originals/S033-sm2-part5-english.pdf | 489,475 bytes | 16 pages

Retrieved UTC: 2026-10-17T18:50:44.241767+00:00

SHA-256: 0947db39f88b81b5e8bc782dcb1c4eded21bfff70aa72e1457c82723d57b6409

S034 · Announcement No.23: release of six cryptographic industry standards

State Cryptography Administration of China (OSCCA)

Document date: 2012-03-21 | Language: Chinese | Format: html

Provenance: Original publisher

Contemporary announcement of industry standards; distinct from later GB/T national standards.

https://oscca.gov.cn/sca/xxgk/2012-03/21/content_1002392.shtml

Preserved: sources/web/S034-oscca-standards-announcement-2012.html | 13,246 bytes

Retrieved UTC: 2026-10-17T18:50:44.307988+00:00

SHA-256: 86b601282319c44cf068df65a3207a7fca7ca968a036134eb80a6993e99da9af

S035 · Official SM4 national-standard record: GB/T 32907-2016

State Administration for Market Regulation / national standards platform

Document date: 2016-08-29 | Language: Chinese | Format: html

Provenance: Original publisher

Official metadata distinguishes issue date from implementation date. A dynamic page snapshot may not preserve all rendered fields.

https://std.samr.gov.cn/gb/search/gbDetailed?id=71F772D81199D3A7E05397BE0A0AB82A

Preserved: sources/web/S035-samr-sm4-record.html | 70,676 bytes

Retrieved UTC: 2026-10-17T18:50:44.456635+00:00

SHA-256: dfb19daab1a72a95006e24d8dc1bd9ec1469b7b891d98a1ecb4b0c4107ede5a5

S036 · Official SM2 Part 1 national-standard record: GB/T 32918.1-2016

State Administration for Market Regulation / national standards platform

Document date: 2016-08-29 | Language: Chinese | Format: html

Provenance: Original publisher

Official record: issue 2016-08-29, implementation 2017-03-01. These dates are not first invention or first classified use.

https://openstd.samr.gov.cn/bzgk/std/newGbInfo?hcno=3EE2FD47B962578070541ED468497C5B

Preserved: sources/web/S036-samr-sm2-record.html | 24,217 bytes

Retrieved UTC: 2026-10-17T18:50:44.572374+00:00

SHA-256: 4933070480f36247e5983d4cc776e58138908a048177231bb11d58a4725769e1

S037 · ZUC Algorithm Specification, Version 1.6

ETSI/SAGE specification hosted by GSMA

Document date: 2011-06-28 | Language: English | Format: pdf

Provenance: Original publisher

Algorithm specification for 128-EEA3 and 128-EIA3. The 2019 upload path is distinct from the 2011 document date.

https://www.gsma.com/solutions-and-impact/technologies/security/wp-content/uploads/2019/05/eea3eia3zucv16

Preserved: sources/originals/S037-zuc-specification-2011.pdf | 293,332 bytes | 18 pages

Retrieved UTC: 2026-10-17T18:50:44.844568+00:00

SHA-256: 07db149438c92b6931eb34708df81134c4fa9a89cb2e0802cb3dd2d59f8907ae

S038 · Der Schluessel M: Verfahren M Allgemein, M.Dv.Nr.32/1

Oberkommando der Kriegsmarine

Document date: 1940 | Language: German | Format: pdf

Provenance: Original wartime manual scan preserved by Crypto Museum

Contemporary naval procedures; the date is not Enigma's invention or first service date.

https://www.cryptomuseum.com/crypto/enigma/files/schluessel_m.pdf

Preserved: sources/originals/S038-enigma-naval-instructions-1940.pdf | 3,416,176 bytes | 17 pages

Retrieved UTC: 2026-10-17T18:50:45.134451+00:00

SHA-256: 93b2f6db52e747cc9d235da5b4589e6516144e749992f46bc368bb07d297de87

S039 · Turing's Treatise on the Enigma, Chapter 1

Alan M. Turing; edited transcription by Ralph Erskine, Philip Marks and Frode Weierud

Document date: circa-1940 | Language: English | Format: pdf

Provenance: Edited transcription of a wartime primary document, preserved by CryptoCellar

Only Chapter 1 is preserved here; it is a transcription with editorial notes, not an unaltered scan of the entire treatise.

https://cryptocellar.org/turing/turchap1.pdf

Preserved: sources/originals/S039-turing-enigma-chapter1.pdf | 36,502 bytes | 10 pages

Retrieved UTC: 2026-10-17T18:50:46.770669+00:00

SHA-256: 4c1525ae8a7b27824d15d50b55b4e40098608fbcb77dff1393d20ff2c7115e28

S040 · General Report on Tunny: Newmanry History portion

Good, Michie, Timms and colleagues; transcription by Tony Sale

Document date: 1945 | Language: English | Format: pdf

Provenance: OCR transcription and reformatting of a primary wartime report

This is the Newmanry portion, reformatted in March 2001, rather than a facsimile of the complete General Report on Tunny. Original page numbers are retained.

https://www.codesandciphers.org.uk/documents/newman/newman.pdf

Preserved: sources/originals/S040-tunny-newmanry-history.pdf | 434,545 bytes | 105 pages

Retrieved UTC: 2026-10-17T18:50:47.348611+00:00

SHA-256: 35d2666628ab70c3f9f4319a9a8e456806f48899efda48ed4cb683116df4a0c1

S041 · SINA im Wandel der Zeit

German Federal Office for Information Security (BSI)

Document date: not verified | Language: German | Format: html

Provenance: Original publisher

Agency retrospective: dates of SINA procurement, migration and system development are distinct from algorithm invention dates.

https://www.bsi.bund.de/DE/Themen/Oeffentliche-Verwaltung/Zulassung/SINA/Historie/historie_node.html

Preserved: sources/web/S041-bsi-sina-history.html | 61,678 bytes

Retrieved UTC: 2026-10-17T18:50:47.968724+00:00

SHA-256: 9e01781d7eda0eab722e01208c1def846c41ee06269eed25b6d2a3147206b3cb

S042 · Abloesung von Chiasmus

German Federal Office for Information Security (BSI)

Document date: 2023-01-25 | Language: German | Format: pdf

Provenance: Original publisher

Version 1.5 is dated 2023-01-25 in the change history. This is migration guidance, not a first algorithm-approval record.

https://www.bsi.bund.de/SharedDocs/Downloads/DE/BSI/Chiasmus/Chiasmus_Abloesung.pdf?__blob=publicationFile&v=6

Preserved: sources/originals/S042-bsi-chiasmus-retirement.pdf | 678,816 bytes | 23 pages

Retrieved UTC: 2026-10-17T18:50:48.037653+00:00

SHA-256: 17235fe949de97659777e63b1dec609f36ef1a8e20b46ca84143354e90951e2f

S043 · RFC 3713: A Description of the Camellia Encryption Algorithm

RFC Editor; named document authors

Document date: 2004-04 | Language: English | Format: text

Provenance: RFC Editor publication; original national standards and RFC dates are distinguished

Preserved in the original publisher format. RFC publication does not itself establish national approval or IETF endorsement; consult the document status statement.

https://www.rfc-editor.org/rfc/rfc3713.txt

Preserved: sources/originals/S043-rfc3713.txt | 25,031 bytes

Retrieved UTC: 2026-10-17T18:50:49.178001+00:00

SHA-256: 126cea1768aed9257fbdf52a4e1938eb15d79c8aacb83d14f60f1320d5b8770f

S044 · RFC 6114: The 128-Bit Blockcipher CLEFIA

RFC Editor; named document authors

Document date: 2011-03 | Language: English | Format: text

Provenance: RFC Editor publication; original national standards and RFC dates are distinguished

Preserved in the original publisher format. RFC publication does not itself establish national approval or IETF endorsement; consult the document status statement.

https://www.rfc-editor.org/rfc/rfc6114.txt

Preserved: sources/originals/S044-rfc6114.txt | 68,618 bytes

Retrieved UTC: 2026-10-17T18:50:49.577540+00:00

SHA-256: d62a3f085cd067c8bdc8b82b0064b097739926385aaad177590285773973a082

S045 · RFC 4269: The SEED Encryption Algorithm

RFC Editor; named document authors

Document date: 2005-12 | Language: English | Format: text

Provenance: RFC Editor publication; original national standards and RFC dates are distinguished

Preserved in the original publisher format. RFC publication does not itself establish national approval or IETF endorsement; consult the document status statement.

https://www.rfc-editor.org/rfc/rfc4269.txt

Preserved: sources/originals/S045-rfc4269.txt | 34,390 bytes

Retrieved UTC: 2026-10-17T18:50:49.832702+00:00

SHA-256: a1132bf03be7ff05ae9be59189632f849a200c2f1c86ce9ecc2f94f51b668f54

S046 · RFC 5794: A Description of the ARIA Encryption Algorithm

RFC Editor; named document authors

Document date: 2010-03 | Language: English | Format: text

Provenance: RFC Editor publication; original national standards and RFC dates are distinguished

Preserved in the original publisher format. RFC publication does not itself establish national approval or IETF endorsement; consult the document status statement.

https://www.rfc-editor.org/rfc/rfc5794.txt

Preserved: sources/originals/S046-rfc5794.txt | 31,049 bytes

Retrieved UTC: 2026-10-17T18:50:50.155704+00:00

SHA-256: e83bdeb56937cb26fc4e3cd6c230ddfa971bc27056c5f202d55820af350cd89d

S047 · Specification of Camellia: a 128-bit block cipher, Version 2.0

NTT and Mitsubishi Electric

Document date: 2001-09-26 | Language: English | Format: pdf

Provenance: Original publisher

Designer specification; edition date is distinct from the earlier 2000 CRYPTREC submission.

https://info.isl.ntt.co.jp/crypt/eng/camellia/dl/01espec.pdf

Preserved: sources/originals/S047-camellia-specification-2001.pdf | 729,102 bytes | 35 pages

Retrieved UTC: 2026-10-17T18:50:50.250209+00:00

SHA-256: 0558d476f05d11da932f2a94b9ff91a84dfe0854d59179b1c21e8d846f7f6074

S048 · SINA: Sichere Inter-Netzwerk Architektur

German Federal Office for Information Security (BSI)

Document date: 2016-01 | Language: German | Format: pdf

Provenance: Original publisher

The imprint states January 2016, BSI-Bro16/322; historical system snapshot.

https://www.bsi.bund.de/SharedDocs/Downloads/DE/BSI/Publikationen/Broschueren/SINA.pdf?__blob=publicationFile

Preserved: sources/originals/S048-bsi-sina-brochure.pdf | 2,365,274 bytes | 44 pages

Retrieved UTC: 2026-10-17T18:50:50.462748+00:00

SHA-256: c44417ea2fe1b3427274c914bf5121fe73e20f36f0462e6767060857300fa093

S049 · The 128-bit Blockcipher CLEFIA: Algorithm Specification, Version 1.0

Sony Corporation

Document date: 2007-06-01 | Language: English | Format: pdf

Provenance: Original publisher

Designer specification; the exact cover date is checked after acquisition.

https://www.sony.net/Products/cryptography/clefia/download/data/clefia-spec-1.0.pdf

Preserved: sources/originals/S049-sony-clefia-specification-2007.pdf | 301,431 bytes | 41 pages

Retrieved UTC: 2026-10-17T18:50:50.949327+00:00

SHA-256: c325748d1923d696b4f0b6c79ca4257eacbaf6cf88ec5265bf28e6842592efe1

S050 · RFC 4357: Additional Cryptographic Algorithms for Use with GOST Algorithms

RFC Editor; named document authors

Document date: 2006-01 | Language: English | Format: text

Provenance: RFC Editor publication; original national standards and RFC dates are distinguished

Preserved in the original publisher format. RFC publication does not itself establish national approval or IETF endorsement; consult the document status statement.

https://www.rfc-editor.org/rfc/rfc4357.txt

Preserved: sources/originals/S050-rfc4357.txt | 114,564 bytes

Retrieved UTC: 2026-10-17T18:50:52.141462+00:00

SHA-256: 384473bcec574b8934e45346006d91efaff7f92bdfd868ec3913cc713f6e21d2

S051 · FIALKA M-125 and M-125-3MN: examination of surviving machines

Tom Perera and David Hamer, Enigma Museum

Document date: 2005 | Language: English | Format: html

Provenance: First-hand artifact photographs and technical examination by the Enigma Museum

The profile uses directly observed construction details. General lineage statements elsewhere on the page are not adopted as verified introduction dates.

https://enigmamuseum.com/mfialka.htm

Preserved: sources/web/S051-fialka-artifact-examination.html | 20,589 bytes

Retrieved UTC: 2026-10-17T18:53:41.239714+00:00

SHA-256: 541cad526eeb037af47c2d1508a80b82d166da909a46cc8cc78d19278fd3a09c

S052 · Specification of Camellia: 2000 CRYPTREC submission

NTT and Mitsubishi Electric

Document date: 2000-07-12 | Language: English | Format: pdf

Provenance: Original publisher

Designer submission dated by NTT's CRYPTREC document index; distinct from later national adoption or an RFC.

https://info.isl.ntt.co.jp/crypt/eng/camellia/dl/cryptrec/00espec.pdf

Preserved: sources/originals/S052-camellia-specification-2000.pdf | 338,418 bytes | 31 pages

Retrieved UTC: 2026-10-17T18:57:13.755407+00:00

SHA-256: cedd73c0e85f4d89c5d7865a20716b455ffc458629c947fec965700185fa22d3

S053 · SMS4 Encryption Algorithm for Wireless Networks

Whitfield Diffie and George Ledin; preserved by Alipay

Document date: 2008-05-15 | Language: English | Format: pdf

Provenance: Translation and technical exposition by Whitfield Diffie and George Ledin, preserved by Alipay

Version 1.03, dated 15 May 2008. Its introduction reports the January 2006 SMS4 issue date.

https://raw.githubusercontent.com/alipay/tls13-sm-spec/master/sm-en-pdfs/sm4/diffie-sm4.pdf

Preserved: sources/originals/S053-diffie-ledin-sms4.pdf | 67,269 bytes | 6 pages

Retrieved UTC: 2026-10-17T18:53:41.242851+00:00

SHA-256: 48f491d978c09d5a902923f412ccdd16aa1ecb7a8440795b7a2fbe48234682a0

S054 · DE416219: Chiffrierapparat

Arthur Scherbius; German patent record

Document date: 1925-07-08 | Language: German | Format: pdf

Provenance: Original patent scan preserved by Crypto Museum

Application/priority date 1918-02-23 and publication date 1925-07-08 must be distinguished; this is not evidence of military approval in 1918.

https://www.cryptomuseum.com/crypto/enigma/patents/files/DE416219.pdf

Preserved: sources/originals/S054-scherbius-patent-de416219.pdf | 716,601 bytes | 6 pages

Retrieved UTC: 2026-10-17T18:53:41.243371+00:00

SHA-256: db77738697c929017682688dbdfed9a9c9134cbec8b384af1bf47e0b5085baa7

S055 · Bletchley Park Cryptographic Dictionary

Government Code and Cypher School; transcription by Tony Sale

Document date: 1944 | Language: English | Format: pdf

Provenance: Transcription of a wartime primary reference, reformatted in 2001

Historical terminology and machine descriptions; transcription and original document dates are kept separate.

https://www.codesandciphers.org.uk/documents/cryptdict/cryptxtt.pdf

Preserved: sources/originals/S055-bletchley-cryptographic-dictionary.pdf | 367,808 bytes | 96 pages

Retrieved UTC: 2026-10-17T18:53:41.667151+00:00

SHA-256: a26ea340f516258ff071eb0732887b1a276535c348742285923d3ad2b0c33e20

S056 · Enigma: Cryptography and war

German Patent and Trade Mark Office (DPMA)

Document date: not verified | Language: English | Format: html

Provenance: Original government agency retrospective

Agency history corroborates the 1918 patent application, distinguishing invention history from later service use. The page date is not treated as an original invention date.

https://www.dpma.de/english/our_office/publications/milestones/computerpioneers/enigma/index.html

Preserved: sources/web/S056-dpma-enigma-history.html | 42,638 bytes

Retrieved UTC: 2026-10-17T18:59:50.363466+00:00

SHA-256: 0346c7099c3007b1adb4308f1fd8e7af8697837afd7f43f563b7d83e137f4d6d